Pair of straight lines

lecture

We know that every linear equation in x and y represents a straight line. That is $Ax + By + C = 0$, where A, B and C are constants, represents a straight line.

The pair of straight lines is in general represented in the following form

$$a\,x^2 + 2m\,xy + b\,y^2 + 2n\,x + 2l\,y + c = 0$$

where c, l, n, b, m, a *are constants.*

1) Condition For General Equation of a Second-Degree Equation (1) to Represent a Pair of Straight Lines

Consider

$$a\,x^2 + 2m\,xy + b\,y^2 + 2n\,x + 2l\,y + c = 0 \quad (1)$$

the necessary condition for equation (1) to represent two straight lines.

is

$$\Delta = \begin{vmatrix} a & m & n \\ m & b & l \\ n & l & c \end{vmatrix} = 0$$

2) Homogeneous Equation of Second Degree in *X* And *Y*

Every homogeneous equation of second degree in x and y represents a pair of straight lines passing through the origin.

Consider the homogeneous equation of second degree in the form $a\,x^2 + 2m\,xy + b\,y^2 = 0$ or $y^2 + \frac{2m}{b}\,xy + \frac{a}{b}\,x^2 = 0$

Suppose that the two straight lines represented by this equation in the form $y = m_1 x \quad , \quad y = m_2 x$

where $m_1\,, m_2$ are the slopes of these two straight lines.

$$(y-m_1x)(y-m_2x) = y^2 + \frac{2m}{b}\,xy + \frac{a}{b}\,x^2 = 0$$

$$\therefore\ y^2 - (m_1 + m_2)\,xy + m_1m_2\,x^2$$

$$= y^2 + \frac{2m}{b}\,xy + \frac{a}{b}\,x^2$$

By equating the coefficients, we find that:

$$m_1 + m_2 = \frac{-2m}{b}\ , \qquad m_1m_2 = \frac{a}{b}$$

3) The Angle Between the Lines Given by Homogeneous Equation

Let θ be the angle between the two straight lines, then

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{\sqrt{(m_1 + m_2)^2 - 4m_1 m_2}}{1 + m_1 m_2}$$

$$= \frac{\sqrt{4\,\frac{m^2}{b^2} - 4\,\frac{a^{\cancel{2}}}{b^{\cancel{2}}}}}{1 + \frac{a}{b}} = \frac{2\sqrt{m^2 - ab}}{a + b}$$

i.e.,

$$\boxed{\tan\theta = \frac{2\sqrt{m^2 - ab}}{a + b}}$$

- If $\theta = \frac{\pi}{2}$, $\longrightarrow$ $\tan\theta = \infty$ $\longrightarrow$ $a + b = 0$
- If $\theta = 0$, $\longrightarrow$ $\tan\theta = 0$ $\longrightarrow$ $\sqrt{m^2 - \mathrm{ab}} = 0$

So,

- The condition that two straight lines are perpendicular is $a + b = 0$
- The condition that two straight lines are parallel $m^2 = \mathrm{ab}$

Solved Examples:

Example

Find the value of k that makes equation

$$k\,x^2 + 2x\,y + y^2 - 1 = 0$$

represent two straight lines and then find them.

Solution

$$a = k\,, b = 1\ \ , c = -1\ \ , m = 1\ \ ,\ \ n = 0\ \ , l = 0$$

$\because$ *the equation represents two straight lines then*

$$\therefore\ \Delta = \begin{vmatrix} a & m & n \\ m & b & l \\ n & l & c \end{vmatrix} = 0$$

$$\therefore \Delta = \begin{vmatrix} k & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & -1 \end{vmatrix} = 0$$

$$\therefore -k + 1 = 0 \quad \Longrightarrow \quad k = 1$$

To find the two lines, we solve the equation as a quadratic equation in x.

$$x^2 + 2\,y\,x + (y^2 - 1) = 0$$

$$A = 1\,,\ \ B = 2\,y\ \ ,C = y^2 - 1$$

$$x = \frac{-B \pm \sqrt{B^2 - 4\,A\,C}}{2\,A}$$

$$= \frac{-2\,y \pm \sqrt{4\,y^2 - 4\,(y^2 - 1)}}{2}$$

$$= -y \pm \sqrt{y^2 - y^2 + 1}$$

$$\therefore x + y = \pm\sqrt{1} = \pm 1$$

$x + y = 1$	$x + y = -1$
$x + y - 1 = 0$	$x + y + 1 = 0$
$x + y - 1 = 0$	$x + y + 1 = 0$

Test your self

Choose the correct answer

1) For what values of k that makes the equation $k x^2 + 2xy + y^2 - 1 = 0$ represents two straight lines,

(a) 1 (b) –1 (c) 2 (d) –3

2)

If the sum of the slopes of the lines given by $x^2 - 2cxy - 7y^2 = 0$ is four times their product, then c has the value

a) 2 (b –1 (c 1 (d –3

thank you